3.7 Monoid Ring

Definition 3.7.1 (Monoid Ring)
Let A be a commutative ring and G be a commutative monoid. We define the ***monoid ring*** *to be the free abelian group*

$$A[G] \coloneqq \bigoplus_{g \in G} A$$

We may write a general element as a formal sum

$$\sum_{g \in G} a_g g$$

where all but finitely many a_g are zero. Multiplication is defined extending the group operation. More precisely

$$\left(\sum_{g \in G} a_g g\right) \cdot \left(\sum_{g \in G} b_g g\right) = \sum_{g \in G} \left(\sum_{(h,k) | h+k=g} a_h b_k g \right)$$

We may verify that the identity is simply e and the multiplication is both associative and distributive.

When G is an abelian group we say that $A[G]$ is the ***group ring****. There is a canonical map $A \to A[G]$ making $A[G]$ into an A-algebra.*

Proposition 3.7.2
Let A, B be commutative rings and G a commutative monoid. Let $\phi : A \to B$ be a ring homomorphism and $\alpha : G \to (B, \times)$ a monoid homomorphism. Then there is a unique homomorphism

$$\phi \ltimes \alpha : A[G] \to B$$

such that $(\phi \ltimes \alpha)(ae) = \phi(a)$ and $(\phi \ltimes \alpha)(1_A g) = \alpha(g)$.

Proof. Define

$$(\phi \ltimes \alpha)\left(\sum_{g \in G} a_g g\right) = \sum_{\substack{g \in G \\ a_g \neq 0}} \phi(a_g)\alpha(g)$$

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